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1. The Natural Numbers

Exercise 1.1

Prove using mathematical induction that for all positive integers n ,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let X be the set of all natural numbers for which the formula above holds true. $1 \in X$ because

$$1 = \frac{1(1+1)}{2}.$$

Now assume that $n > 1$, and that $k \in X$ for all $k < n$. We know from our induction hypothesis that the formula holds for $n-1$, so

$$1 + 2 + 3 + \cdots + (n-1) = \frac{(n-1)n}{2}.$$

We add n to both sides to obtain

$$\begin{aligned} 1 + 2 + 3 + \cdots + (n-1) + n &= \frac{(n-1)n}{2} + n \\ &= \frac{n^2 - n + 2n}{2} \\ &= \frac{n^2 + n}{2} \\ &= \frac{n(n+1)}{2}. \end{aligned}$$

Therefore, $n \in X$. It follows from the principle of mathematical induction that $X = \mathbb{N}$. □

Exercise 1.2

Prove using mathematical induction that for all positive integers n ,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(2n+1)(n+1)}{6}.$$

Proof. Let X be the set of natural numbers from which the formula above holds. $1 \in X$ because

$$1^2 = \frac{1(2 \cdot 1 + 1)(1 + 1)}{6}.$$

Now suppose that $n > 1$, and that $k \in X$ for all $k < n$. In particular, $n-1 \in X$ so

$$1^2 + 2^2 + \cdots + (n-1)^2 = \frac{n(n-1)(2n-1)}{6}.$$

Adding n^2 to both sides, we deduce that

$$\begin{aligned}
 1^2 + 2^2 + \cdots + (n-1)^2 + n^2 &= \frac{n(n-1)(2n-1)}{6} + n^2 \\
 &= \frac{2n^3 - 3n^2 + n + 6n^2}{6} \\
 &= \frac{2n^3 + 3n^2 + n}{6} \\
 &= \frac{n(2n^2 + 3n + 1)}{6} \\
 &= \frac{n(2n+1)(n+1)}{6}.
 \end{aligned}$$

So $n \in X$, which closes the induction. $\square$

Exercise 1.3

Assuming the triangle inequality ($|x+y| \leq |x|+|y|$) is true, show that $|x_1+x_2+\cdots+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$ for all $n \in \mathbb{N}$.

Proof. Let X be the set of all natural numbers n such that $|x_1+x_2+\cdots+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$. $1 \in X$ and it's given that $2 \in X$. Suppose that $k \in X$ for all $k < n$ where $n > 2$. In particular, that means $n-1 \in X$. Therefore, $|x_1+x_2+\cdots+x_{n-1}| \leq |x_1|+|x_2|+\cdots+|x_{n-1}|$. We add $|x_n|$ to both sides to obtain

$$|x_1+x_2+\cdots+x_{n-1}|+|x_n| \leq |x_1|+|x_2|+\cdots+|x_{n-1}|+|x_n|.$$

Let $y = x_1+x_2+\cdots+x_{n-1}$, then the left-hand side of the inequality above becomes $|y|+|x_n|$. Because $2 \in X$, we know $|y+x_n| \leq |y|+|x_n|$. Therefore, $|y+x_n| \leq |x_1|+|x_2|+\cdots+|x_n|$, which closes the induction and shows that $X = \mathbb{N}$. $\square$

Exercise 1.4

Given a positive integer n , recall that $n! = 1 \cdot 2 \cdot 3 \cdots n$ (this is read as n **factorial**). Provide an inductive definition for $n!$. (It is customary to actually use this definition at $n = 0$, setting $0! = 1$.)

Proof.

$$n! = n(n-1)!$$

$\square$

Exercise 1.5

Prove that $2^n < n!$ for all $n \geq 4$.

Proof. Let X be the set of all natural numbers n such that $2^n < n!$. Define $S := X \cup \{1, 2, 3\}$, we show that $S = \mathbb{N}$ by induction. By definition, $1 \in S$. Now suppose that $k \in S$ for all $k < n$ where $n > 1$. If $n = 2$ or $n = 3$, then by definition $n \in S$. If $n = 4$, then $2^4 = 16$ and $4! = 24$. Clearly $16 < 24$, therefore $n \in X$. Now we can solely focus on the cases where $n > 4$. We know $n - 1 \in S$, so $2^{n-1} < (n - 1)!$ or $n - 1 \in \{1, 2, 3\}$. But $n > 4$ so $n - 1 \notin \{1, 2, 3\}$. So it must be the case that $2^{n-1} < (n - 1)!$. $\square$